

Q1 Which Optimization produced the greatest improvement?

Ans: Implementing the ThreadPoolExecutor produced the greatest improvement by efficiently managing resources, which prevented memory exhaustion and significantly increased message throughput compared to the previous unbounded thread creation.

Q2 Why is scalability important?

Scalability ensures the server remains responsive and stable as the number of concurrent clients increases, preventing system crashes, dropped connections, and unacceptable network delays under heavy load.

Q3 What reliability issue did you discover?

I discovered that sudden client disconnections or network drops previously left 'zombie' threads and locked server resources. This was resolved by adding automatic disconnection and proper exception handling to clean up active sessions gracefully.

Q4 Which optimizations was the most difficult?

Managing thread safety and ensuring resources were properly released during simultaneous client disconnections was the most difficult as it required careful lock management to avoid deadlocks or crashing the main server loop.

Q5 What additional improvements are required to support 100 users?

Aside from supporting 100 concurrent users, the application would need to transition from standard threading to asynchronous I/O and replace the hard CSV storage with robust database.